Starbucks Mobile App Usability Test

A task-based usability evaluation of the Starbucks iOS mobile app, focused on drink customization, card reload, and store locator flows.

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Academic UX Case Study

Note: This case study was completed as part of CCPS 613: Human-Computer Interaction and has been reformatted from the original lab submission into a portfolio-style UX report.

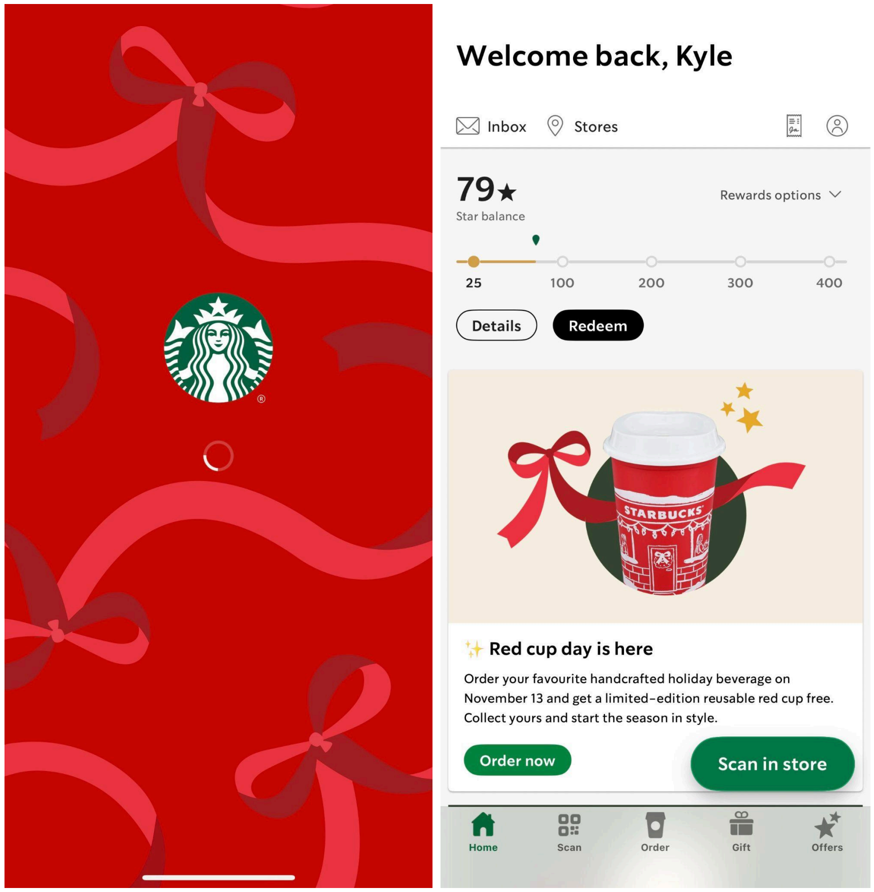
Executive Summary

This usability test evaluated the Starbucks mobile app with five participants representing a range of ages, device types, smartphone confidence levels, and Starbucks familiarity. Three core tasks were tested: placing a drink order, navigating the reload flow, and finding a nearby store. Quantitative metrics included task completion rates, time on task, and errors. Qualitative observations captured verbal feedback, behavioural cues, and emotional responses. A post-test SUS questionnaire was administered to all participants.

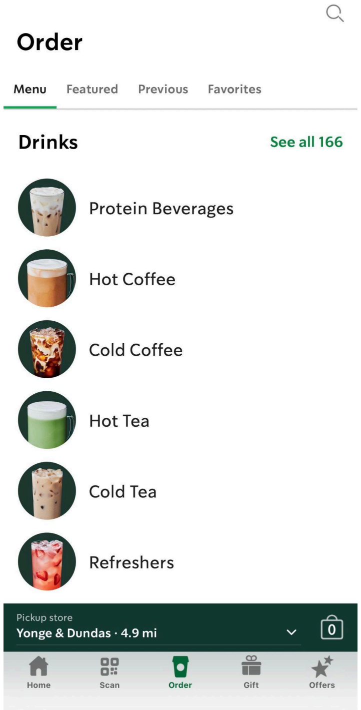
Overall, the Starbucks app performed reasonably well, particularly for confident users, but notable usability issues emerged in customization, payment and reload flows, and store locator navigation. The combined SUS score of 64.5 places the system slightly below the industry benchmark of 68, indicating that while the app is usable, several design improvements would meaningfully enhance user experience.

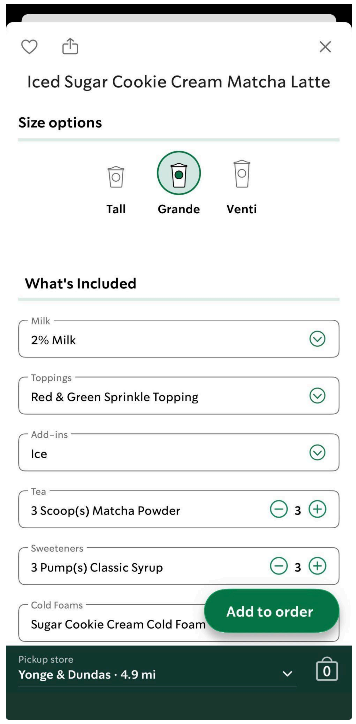
1. Project Overview

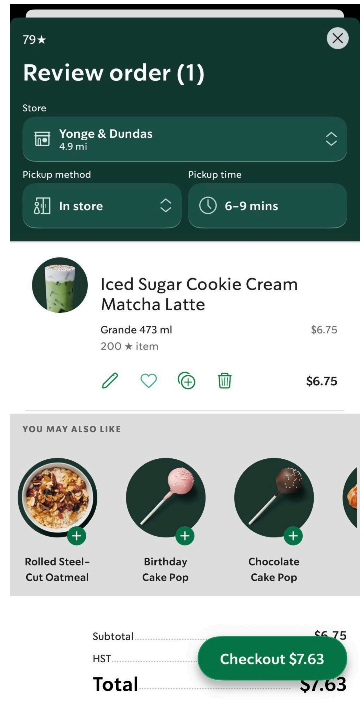
- **System Selection:** Starbucks Mobile App (iOS)
    - This usability test evaluated the Starbucks mobile app on iOS. The app enables users to browse the menu, customize drinks, place mobile orders, reload a Starbucks Card, access rewards, and locate nearby stores. The evaluation focused on three common user flows: ordering and customizing a drink, reloading a Starbucks Card, and finding a nearby store.



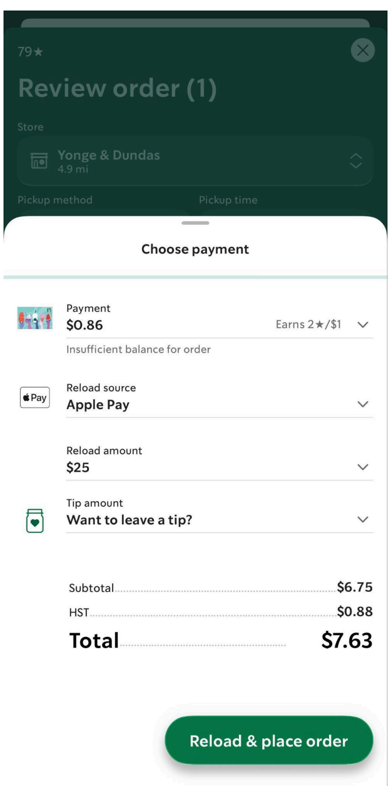
- **Core Features Evaluated**
  - Feature 1: Mobile Ordering & Drink Customization
    - Users browse the menu, select an item, and customize ingredients, size, and add ons before placing an order.
  - Feature 2: Starbucks Card Reload
    - Users can add funds to their Starbucks Card using saved payment methods or Apple Pay.
  - Feature 3: Store Locator
    - Users can locate nearby Starbucks stores and view hours, amenities, and directions.
- **Test Tasks**
  - Task 1 – Order and Customize a Drink
    - “Order a Grande Iced Latte and customize it by selecting almond milk and adding two pumps of vanilla syrup. Proceed to the checkout screen but do not complete the purchase.”

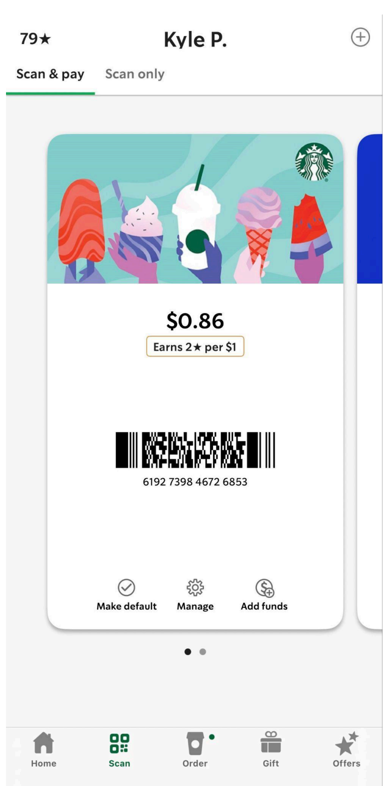


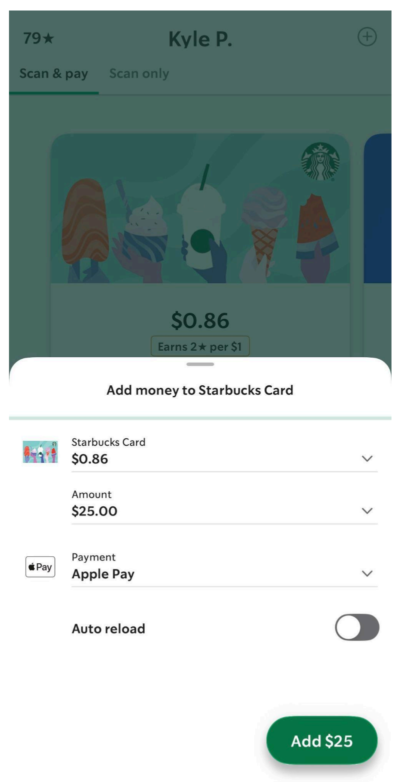




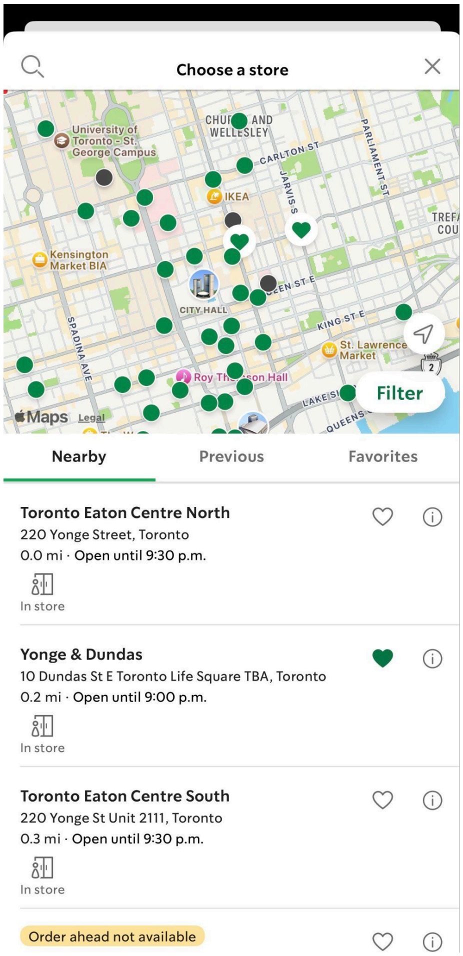
- Task 2 – Reload Starbucks Card
  - "Reload your Starbucks Card with ten dollars using Apple Pay. Stop when you reach the confirmation screen without finalizing the reload."

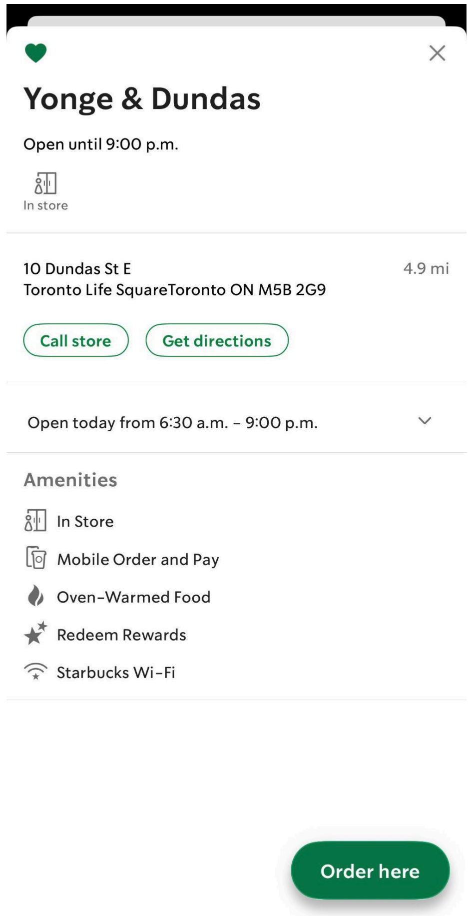






- Task 3 – Locate the Nearest Store
    - “Find the nearest Starbucks location to your current position and view the store’s hours and amenities.”



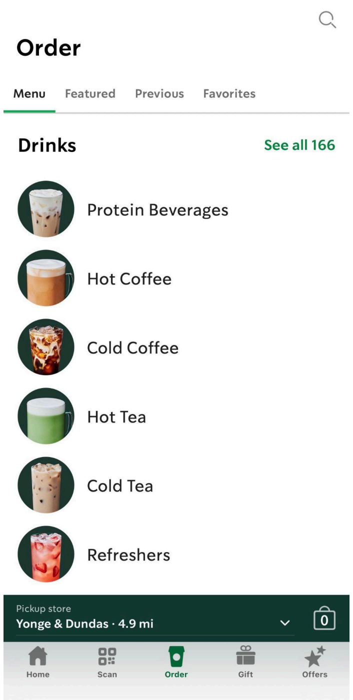


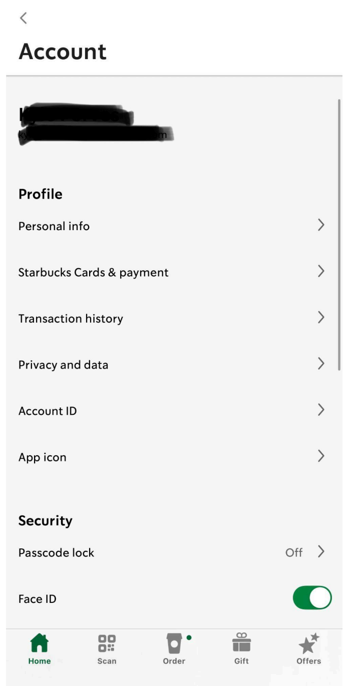
2. Evaluation Context

- ### User Characteristics Considered
    - Age Range (18 to 55):
        - Mobile apps rely on quick visual recognition, gesture precision, and familiarity with smartphone conventions. Age can influence speed, comfort, and error frequency.
    - Smartphone Experience Level (Novice, Intermediate, Experienced):
        - Interaction efficiency is directly affected by how comfortable users are with navigation, tapping accuracy, scrolling, and using mobile menus.

- Experience With Starbucks App or Similar Ordering Apps:
  - Prior knowledge of mobile food ordering can significantly impact effectiveness, efficiency, and satisfaction. Users who have never ordered through a mobile app may struggle more with customization or payment flows.
- Frequency of Mobile App Usage (Daily, Weekly, Rarely):
  - Frequency influences mental models and expectations of where features should be located.
- Payment Experience (Apple Pay or Mobile Wallet Familiarity):
  - Since one task involves reloading a Starbucks card, familiarity with mobile payment affects speed and confidence.

- **Environmental Variables Considered**
  - Device Type and Screen Size (iPhone, standard size screen):
    - Screen dimensions affect visibility of menus, readability of customization lists, and perceived complexity.
  - Input Method (Touchscreen, one handed use):
    - One handed scrolling, tapping, and gesture use can increase slips, missed taps, or difficulty accessing buttons located near screen edges.
  - Lighting Conditions (Normal indoor lighting):
    - Poor lighting impacts readability and can slow task completion. Indoor lighting simulates typical usage scenarios such as ordering from home or indoors.
  - Internet Connection Quality (Stable Wi Fi):
    - A reliable connection reduces confounding factors such as loading delays, ensuring performance differences are due to usability rather than network issues.
  - Noise and Distractions (Minimal distractions):
    - Low distraction environments allow users to focus on navigation and reduce external factors that could affect task performance.
  - Time Pressure (Casual pace, no urgency):
    - The absence of time stress mirrors typical mobile ordering behavior where users browse and customize at their own pace. Time pressure would artificially affect errors and satisfaction.





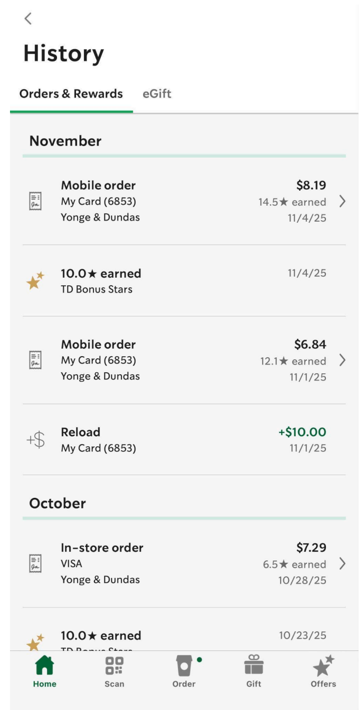


Figure 1: The customization interface on iPhone, where screen size affects information density and requires scrolling to access all options. This contextual factor directly impacts efficiency metrics, as users must navigate through multiple collapsible sections to complete drink customization.

Figure 2 and 3: Touch targets and button placement throughout the app interface, where one-handed interaction can affect tap accuracy and increase the likelihood of slips, particularly for elements positioned near screen edges or requiring precise finger movements.

3. Participants

- Participant 1 – Work Colleague
  - Age: 33
  - Smartphone Experience: Very experienced
  - Starbucks App Experience: Weekly user
  - Mobile Payment Use: Uses Apple Pay frequently
  - Environment: Office desk, quiet
  - Device: iPhone 17 Pro

- Participant 2 – Partner
    - Age: 35
    - Smartphone Experience: Experienced
    - Starbucks App Experience: Regular user (monthly or more)
    - Mobile Payment Use: Uses Apple Pay daily
    - Environment: Living room, calm lighting
    - Device: iPhone 15 Pro Max

- Participant 3 – Student (Young Adult)
    - Age: 21
    - Smartphone Experience: Very experienced
    - Starbucks App Experience: Rare user, first time customizing drinks
    - Mobile Payment Use: Occasionally uses Apple Pay
    - Environment: Campus lounge, bright lighting
    - Device: iPhone 16

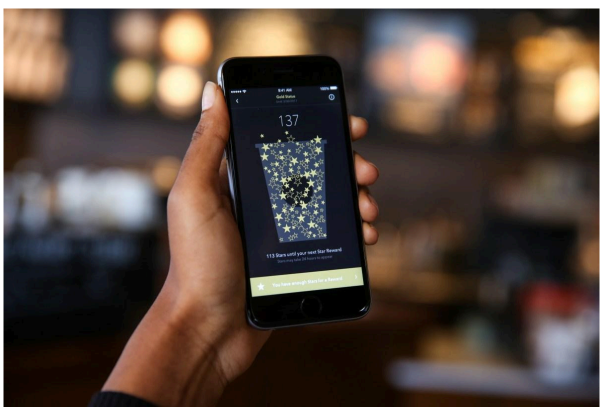
- Participant 4 – Mother
    - Age: 65
    - Smartphone Experience: Intermediate
    - Starbucks App Experience: Occasional user
    - Mobile Payment Use: Does not use Apple Pay; prefers physical card
    - Environment: Dining room, bright daytime light
    - Device: iPhone 14

- Participant 5 – Father
    - Age: 67
    - Smartphone Experience: Low–intermediate
    - Starbucks App Experience: Rare user, usually orders in person
    - Mobile Payment Use: Does not use mobile wallets
    - Environment: Living room, evening lighting
    - Device: iPhone 13

Participant Distribution Summary

Criteria	Range Represented
Age	21-67 (spanning 4 decades)
Experience	Low-intermediate to Very experienced
App Familiarity	Rare users to Weekly users
Payment Methods	No mobile wallets to Daily Apple Pay users
Devices	iPhone 13 through iPhone 17 Pro
Environments	Office, home, campus (varied lighting)

The participant group represented a range of ages, smartphone confidence levels, Starbucks app familiarity, payment comfort, devices, and environments. This allowed the test to compare how confident and less confident mobile users experienced the same core flows.



4. Methodology

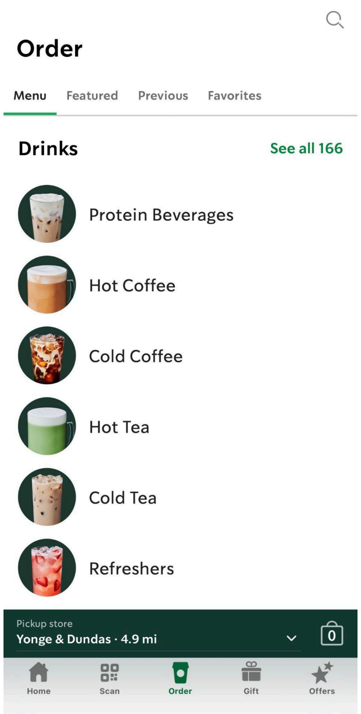
Participants completed three structured tasks while thinking aloud. I observed each session and recorded task completion, time on task, errors, click-path notes, verbal feedback, behavioral cues, and emotional responses. After completing the tasks, participants filled out the System Usability Scale (SUS) questionnaire to measure perceived usability and satisfaction.

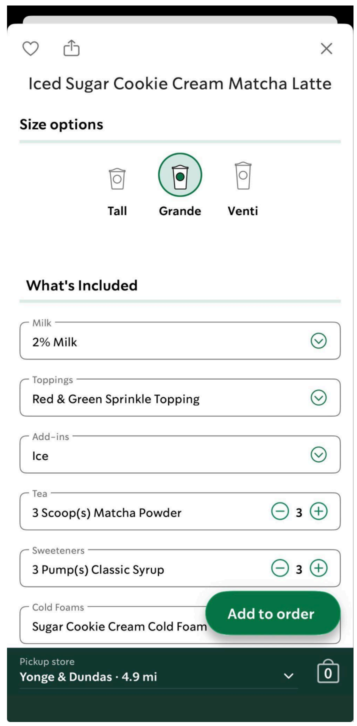
A. Quantitative Metrics (Effectiveness and Efficiency)

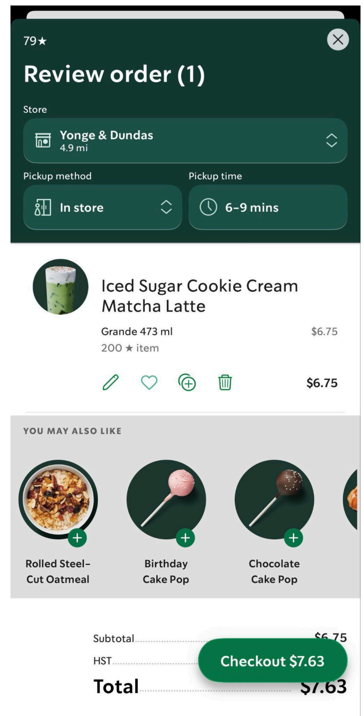
User	Task	Completion (Y/N/Assisted)	Time (sec)	Errors (Slips / Mistakes)	Click Path Notes
P1	Task 1	Y	32	1 slip	Missed the “Customize” button once before noticing it.
P1	Task 2	Y	18	0	Straightforward reload; used Apple Pay instantly.
P1	Task 3	Y	22	0	Found nearest store quickly using the map view.
P2	Task 1	Y	45	1 mistake	Accidentally chose whole milk before switching to almond milk.
P2	Task 2	Y	28	0	Hesitated before confirming the amount.
P2	Task 3	Y	35	0	Scrolled past the nearest store once before tapping on it.
P3	Task 1	Assisted	60	2 slips	Didn’t realize customization required scrolling; tapped back once by accident.
P3	Task 2	N	40	1 mistake	Unsure how to use Apple Pay; stopped before final screen.
P3	Task 3	Y	30	0	Used search bar instead of the map view.
P4	Task 1	Y	55	1 slip	Struggled to find syrup options at first.
P4	Task 2	N	42	1 mistake	Didn’t want to proceed without using a physical card.
P4	Task 3	Y	48	0	Took extra time zooming into the map.
P5	Task 1	Assisted	70	2 mistakes	Unsure how to customize the drink; selected wrong size twice.

P5	Task 2	N	50	1 slip	Tapped “Reload” but backed out due to confusion.
P5	Task 3	Y	40	0	Eventually found store details after scanning the list.

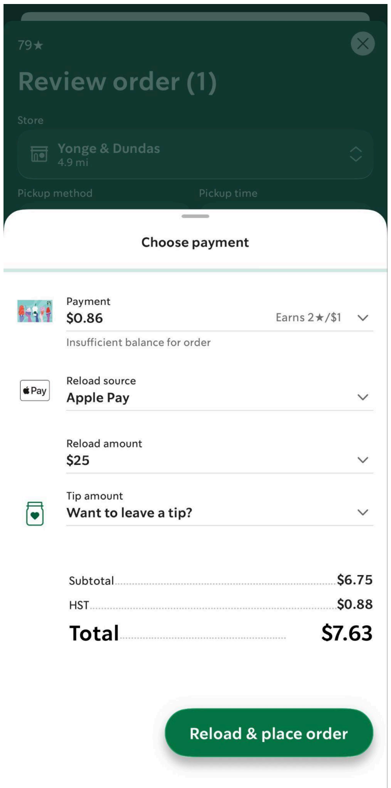
- Task 1 – Order and Customize a Drink

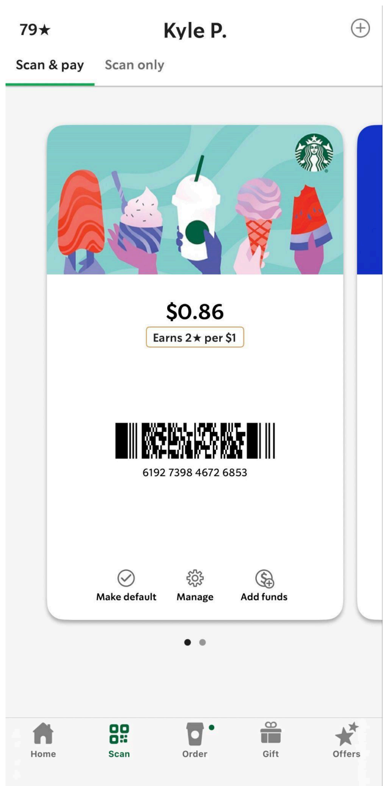


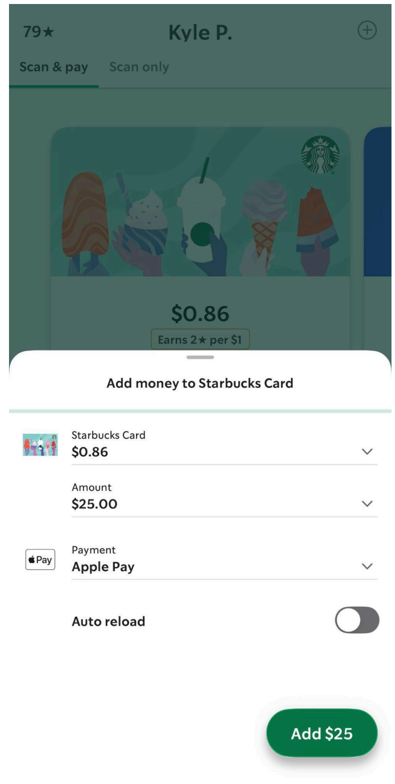




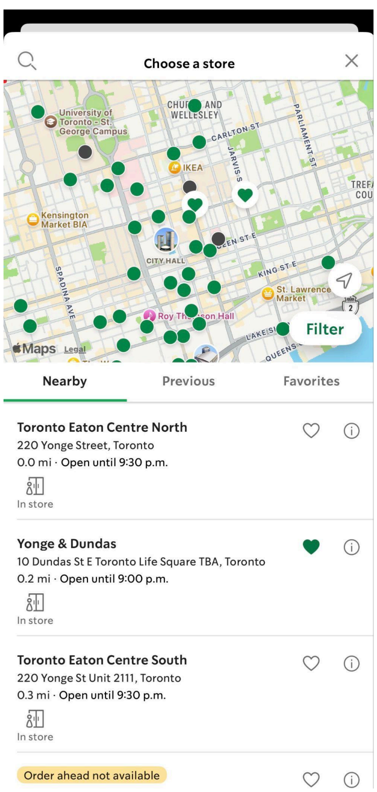
- Task 2 – Reload Starbucks Card

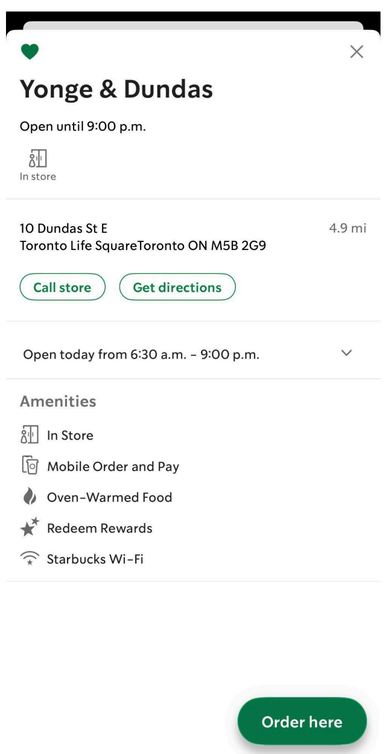






- Task 3 – Locate the Nearest Store





B. Qualitative Observations

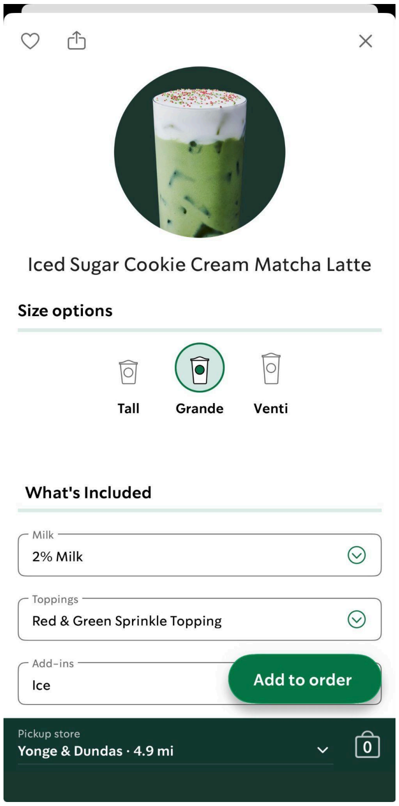
- Participant 1: Work Colleague (iPhone 17 Pro)
  - Task 1
    - Verbal: “Oh, the customize button is tiny. Almost missed it.”
    - Behavioural: Quick scrolling, slight pause before selecting almond milk.
    - Emotional: Neutral, mild concentration.
  - Task 2
    - Verbal: “Easy, I always do this.”
    - Behavioural: Immediate tap on Apple Pay; no hesitation.
    - Emotional: Confident, relaxed.
  - Task 3
    - Verbal: “Map loaded fast. Good.”
    - Behavioural: Pinched to zoom once, tapped first store icon.
    - Emotional: No frustration; smooth experience.

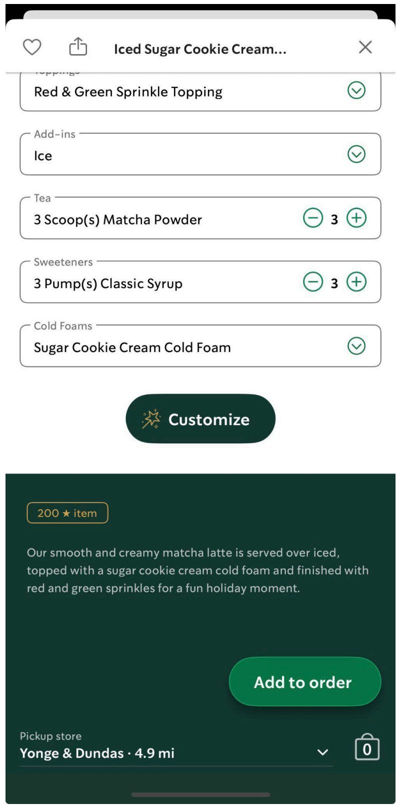
- Participant 2: Partner (iPhone 15 Pro Max)
  - Task 1
    - Verbal: “Wait, where’s the milk option… oh found it.”
    - Behavioural: Backed up once after picking the wrong milk.
    - Emotional: Light annoyance for two seconds; then fine.
  - Task 2
    - Verbal: “I always worry I’ll reload too much by accident.”
    - Behavioural: Hovered over the dollar amounts before choosing.
    - Emotional: Slight caution, but not frustrated.
  - Task 3
    - Verbal: “I saw the store but scrolled past it.”
    - Behavioural: Slow upward scrolling; tapped the wrong store first.
    - Emotional: Minor sigh, then back to normal.

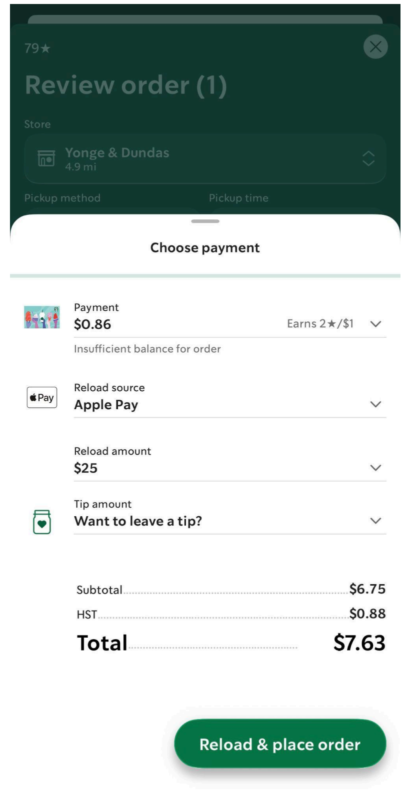
- Participant 3: Student (iPhone 16)
  - Task 1
    - Verbal: “Oh wow, there’s more options down here? I didn’t notice.”
    - Behavioural: Scrolled up and down twice; tapped back accidentally.
    - Emotional: Small laugh at their own confusion.
  - Task 2
    - Verbal: “I don't use Apple Pay... um, is this... I'm not sure what to press.”
    - Behavioural: Stared at the screen for a long time; hesitated.
    - Emotional: Uncertain; eventually gave up.
  - Task 3
    - Verbal: “I’ll just type the address instead.”
    - Behavioural: Used search bar instead of map; steady interaction.
    - Emotional: Comfortable once they switched to the search field.

- Participant 4: Mother (iPhone 14)
  - Task 1
    - Verbal: “There are a lot of things here… where do I change the syrup?”
    - Behavioural: Slow scrolling; tapped into the wrong customization section first.
    - Emotional: Mild frustration but recovered.
  - Task 2
    - Verbal: “I don’t want to use my phone for payments.”
    - Behavioural: Pressed reload, then backed out immediately.
    - Emotional: Low confidence, discomfort with the payment flow.
  - Task 3
    - Verbal: “Let me zoom in a bit… okay I see it now.”
    - Behavioural: Pinched to zoom several times; took longer to tap the correct store.
    - Emotional: Calm; slightly focused.

- Participant 5: Father (iPhone 13)
  - Task 1
    - Verbal: “Wait, why did it... I thought I had Grande selected. Why did it switch sizes on me? Did I tap something?”
    - Behavioural: Accidentally changed the drink size twice; slow navigation.
    - Emotional: Small sigh; mild irritation.
  - Task 2
    - Verbal: “I’m not sure what this wants me to do.”
    - Behavioural: Tapped reload, hesitated, then cancelled.
    - Emotional: Uncertain, avoided the task.
  - Task 3
    - Verbal: “Found it… eventually.”
    - Behavioural: Scrolled through list view; took time reading store names.
    - Emotional: Neutral but slightly fatigued.







C. System Usability Scale Results

SUS Item / Participant	P1	P2	P3	P4	P5
Q1	4	4	3	3	3
Q2	2	2	2	2	3
Q3	5	4	4	3	3
Q4	2	2	3	3	3
Q5	4	4	3	3	3
Q6	2	2	2	2	3
Q7	5	4	3	3	3
Q8	2	2	3	3	3
Q9	4	4	3	3	3

Q10	2	2	2	2	3
Final SUS Score	**80.0**	**75.0**	**60.0**	**57.5**	**50.0**

The average SUS score across all five participants was 64.5. This falls slightly below the commonly used benchmark of 68, suggesting that the Starbucks app is usable but has noticeable friction points, particularly for users who are less confident with mobile payments or app-based customization.

5. Key Findings and Analysis

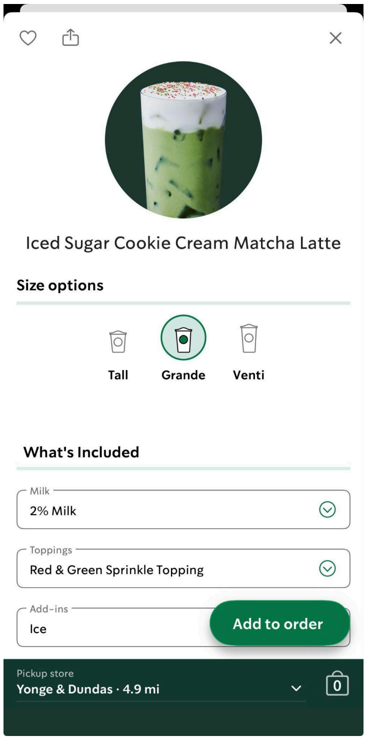
Across the three tasks, several recurring patterns emerged:

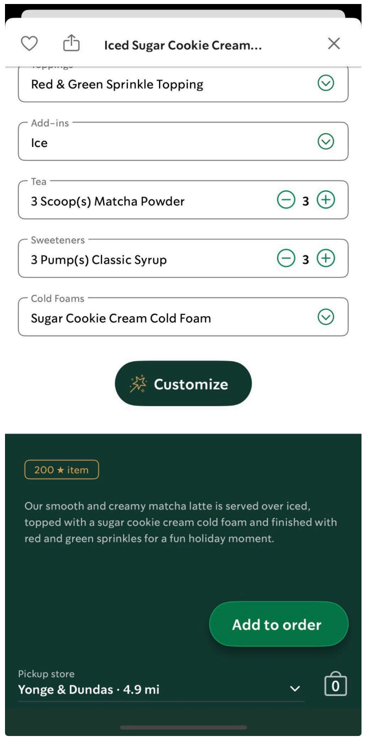
- Multiple users had difficulty locating or noticing the customization entry point.
- The payment and reload flow caused uncertainty, especially for less confident smartphone users.
- The store locator interface led to scanning and selection errors due to ambiguous map icons and crowded visuals.
- The customization screen itself felt overloaded, impacting scanning and speed.
- Minor inconsistencies in screen layout caused unnecessary hesitation for less experienced users.

These issues affected all three usability pillars: Effectiveness, Efficiency, and Satisfaction.

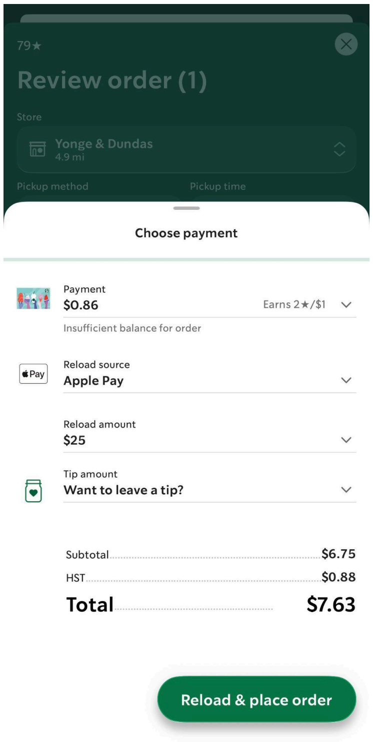
A. Identifying Usability Problems

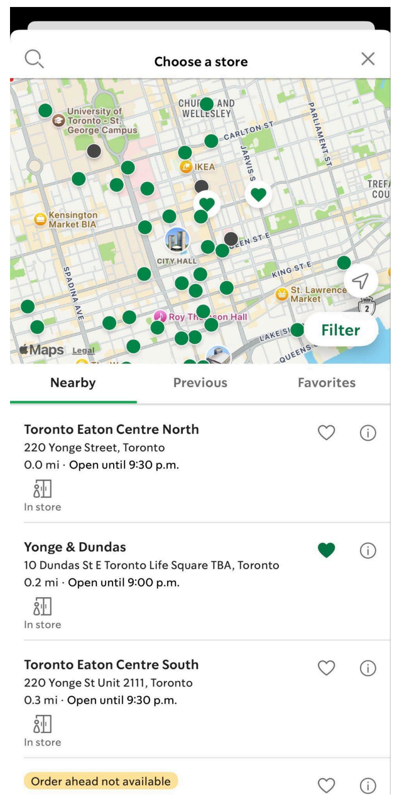
- Usability Problem 1: Customization Options Are Hard to Notice and Easy to Miss
    - The "Customize" option is positioned below the fold and requires scrolling to discover, while the prominent "Add to Order" button remains visible at all times. This layout causes users to either miss the customization feature entirely (P1: "Almost missed it") or assume customization is not available. Multiple participants (P1, P2, P3) experienced delays or errors locating this critical function, directly impacting Efficiency and creating brief Satisfaction drops as users questioned whether drink modification was possible.



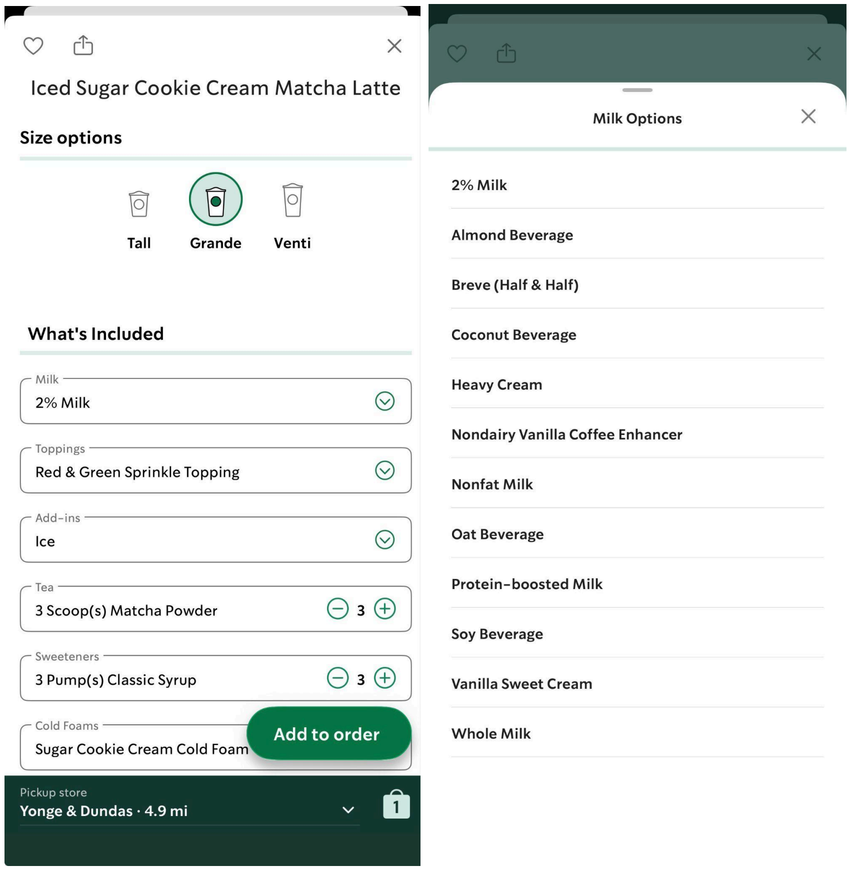


- Usability Problem 2: Payment and Reload Flow Causes Uncertainty for Less Confident Users
    - The reload screen automatically defaults to a $25 amount without clear visual emphasis that this value is modifiable. While users can adjust the amount in $5 increments, the pre-filled value is not prominently highlighted as changeable. This design choice caused hesitation (P2: "I always worry I'll reload too much by accident"), confusion about how to modify the amount, and contributed to task abandonment for three participants (P3, P4, P5) who were uncertain about proceeding. This issue severely impacted both Effectiveness (60% task failure rate) and Satisfaction (lower confidence scores on SUS Q9).





- Usability Problem 3: Store Locator Icons and List View Are Not Immediately Clear
    - The store locator presents multiple icons with similar appearances, and switching between map and list views is not clearly distinguished, leading to confusion and mis-taps.
- Usability Problem 4: Too Many Options on Customization Screen Create Cognitive Overload
    - The customization interface shows many collapsible sections at once, causing scanning difficulty and forcing inexperienced users to scroll excessively.



- Usability Problem 5: Inconsistent Interaction Patterns Across Screens
    - Navigation patterns differ between ordering, payments, and store search screens. Inconsistencies in layout and button placement lead to slower task completion and erode user confidence.

B. Issue Prioritization

Severity was assigned based on impact on task completion, frequency across participants, and effect on user confidence.

Usability Problem	Impact	Frequency	Severity Score	Severity Rank
1. Customization options are hard to notice	Medium	High	3	Serious
2. Payment and reload flow causes uncertainty	High	Medium	5	Critical

3. Store locator icons and views are confusing	Medium	Medium	3	Serious
4. Customization screen feels overloaded	Medium	High	3	Serious
5. Inconsistent interaction patterns slow users down	Low to Medium	Medium	1	Minor

C. Evidence and Triangulation

Problem 1: Customization options are hard to notice and easy to miss

Triangulated Evidence:

- Quantitative: Multiple users spent longer than expected on Task 1 and made slips while selecting customization options. P1 paused, P2 backed out once, and P3 scrolled repeatedly through the menu.
- Qualitative Quotes:
    - “Oh, the customize button is tiny. Almost missed it.” (P1)
    - “Wait, where’s the milk option… oh found it.” (P2)
- Behavioral Cues: Frequent scrolling, backtracking, and hesitation.
- SUS Indicators: Lower confidence and higher perceived complexity for mid-scoring users (Q2, Q6, Q8).
- Interpretation: Users struggled to notice the customization entry point, which affected Efficiency and caused a brief drop in Satisfaction.

Problem 2: Payment and reload flow causes uncertainty for less confident users

Triangulated Evidence:

- Quantitative: Several users failed to complete Task 2. P3, P4, and P5 did not finish the reload process.
- Qualitative Quotes:
    - “I don’t use Apple Pay. Not sure what to press.” (P3)
    - “I always worry I’ll reload too much by accident.” (P2)
    - “I don’t want to use my phone for payments.” (P4)
- Behavioral Cues: Users hovered over options, backed out of the screen, or cancelled the action entirely.
- SUS Indicators: Lower confidence (Q9) and higher perceived complexity (Q2, Q4).
- Interpretation: This issue directly affected Effectiveness because several users could not complete the task. It also had the strongest negative impact on Satisfaction.

Problem 3: Store locator icons and view modes create confusion

Triangulated Evidence:

- Quantitative: Task 3 took longer for several users, and some made accidental taps.
- Qualitative Quotes:
  - “I saw the store but scrolled past it.” (P2)
  - “Let me zoom in a bit… okay I see it now.” (P4)
- Behavioral Cues: Repeated zooming, slower scanning, and tapping the wrong store icon.
- SUS Indicators: Medium scores on confidence and ease of use (Q3, Q9).
- Interpretation: Unclear differences between map and list views affected Efficiency and created minor Satisfaction issues.

Problem 4: The customization screen feels overloaded and hard to scan

Triangulated Evidence:

- Quantitative: Increased scrolling time during Task 1 for P3, P4, and P5.
- Qualitative Quotes:
  - “There are a lot of things here… where do I change the syrup?” (P4)
  - “Oh wow, there’s more options down here?” (P3)
- Behavioral Cues: Accidental back navigation and repeated up and down scrolling.
- SUS Indicators: Items reflecting perceived inconsistency and complexity (Q2, Q6, Q8).
- Interpretation: This issue primarily affects Efficiency because users must sort through many stacked sections before making a choice.

Problem 5: Inconsistent interaction patterns slow users down

Triangulated Evidence:

- Quantitative: Timing varied across screens. Ordering tasks were relatively quick, while payment tasks took longer, especially for older users.
- Qualitative Quotes:
  - “Why did it switch sizes on me?” (P5)
  - “I’m not sure what this wants me to do.” (P5, Task 2)
- Behavioral Cues: Extra taps, brief moments of confusion, and general uncertainty when switching between sections of the app.
- SUS Indicators: Moderate scores on integration and predictability (Q5, Q6).
- Interpretation: This issue did not block task completion, but it reduced Efficiency for users who were less familiar with mobile apps.

D. SUS Score Interpretation

The individual SUS scores for all five participants were 80.0, 75.0, 60.0, 57.5, and 50.0. Adding these values and dividing by the number of participants produces a final system SUS score:

(80.0 + 75.0 + 60.0 + 57.5 + 50.0) / 5 = 64.5

The accepted SUS benchmark is approximately 68. Systems that fall below this value are considered to have noticeable usability issues, while scores above it indicate generally good usability. With an overall SUS score of 64.5, the Starbucks app lands just below the benchmark. This suggests that the app is usable for many users but still presents meaningful friction points that affect confidence and ease of use, especially for less experienced or less tech-confident participants. This aligns with the qualitative findings and the observed difficulties in customization, the reload process, and navigating the store locator.

6. Recommendations

Each recommendation is directly linked to a documented usability issue and supported by triangulated evidence from the study.

Recommendation 1: Make customization more visible

Linked issue: Customization options are hard to notice

Suggested change: Move the Customize option above the fold, increase its visual weight, or add a clear customization prompt before checkout.

Expected impact: Improves efficiency and reduces missed customization actions

Recommendation 2: Clarify the reload flow

Linked issue: Payment and reload flow causes uncertainty

Suggested change: Make the reload amount easier to edit, visually emphasize that the default amount can be changed, and add clearer confirmation language before the user commits to the reload.

Expected impact: Improves task completion and user confidence

Recommendation 3: Improve store locator clarity

Linked issue: Store locator icons and list view are confusing

Suggested change: Differentiate map icons more clearly, improve the visual distinction between map and list views, and highlight the nearest location more prominently.

Expected impact: Reduces scanning errors and improves speed

Recommendation 4: Reduce visual clutter in customization

Linked issue: Customization screen feels overloaded

Suggested change: Group related customization options more clearly, collapse lower-priority options by default, and use stronger section labels for common choices such as milk, syrup, size, and toppings.

Expected impact: Improves scanning and decision-making

Recommendation 5: Standardize interaction patterns

Linked issue: Inconsistent navigation and button placement

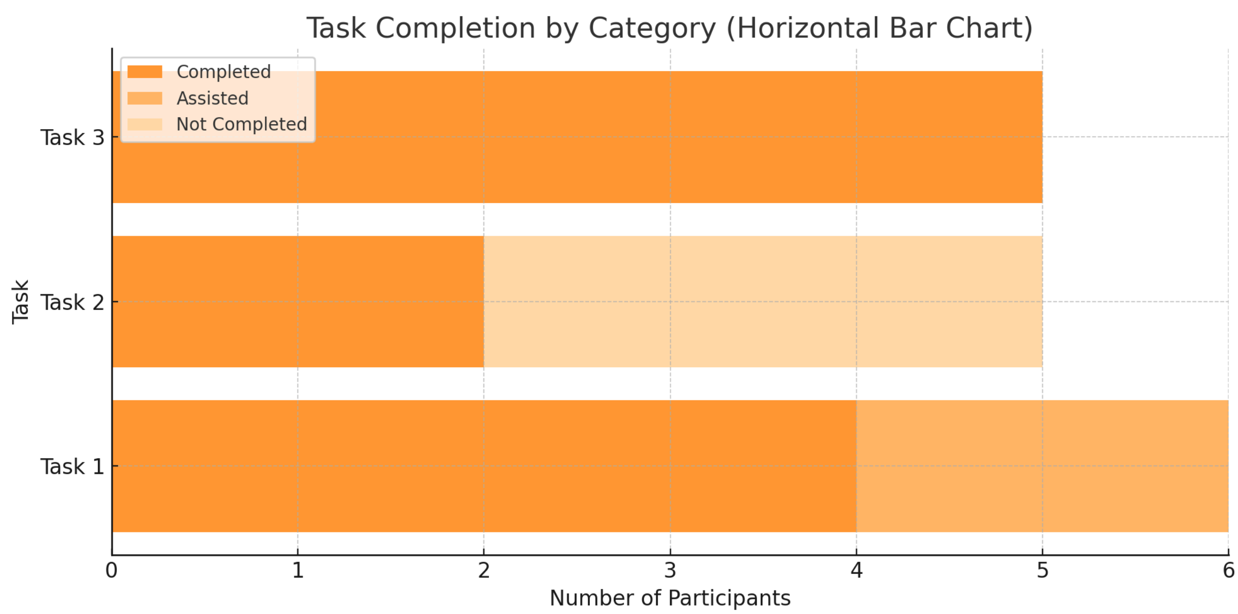
Suggested change: Align button placement, navigation behavior, and confirmation patterns across ordering, reload, and store locator screens so users encounter more predictable interactions.

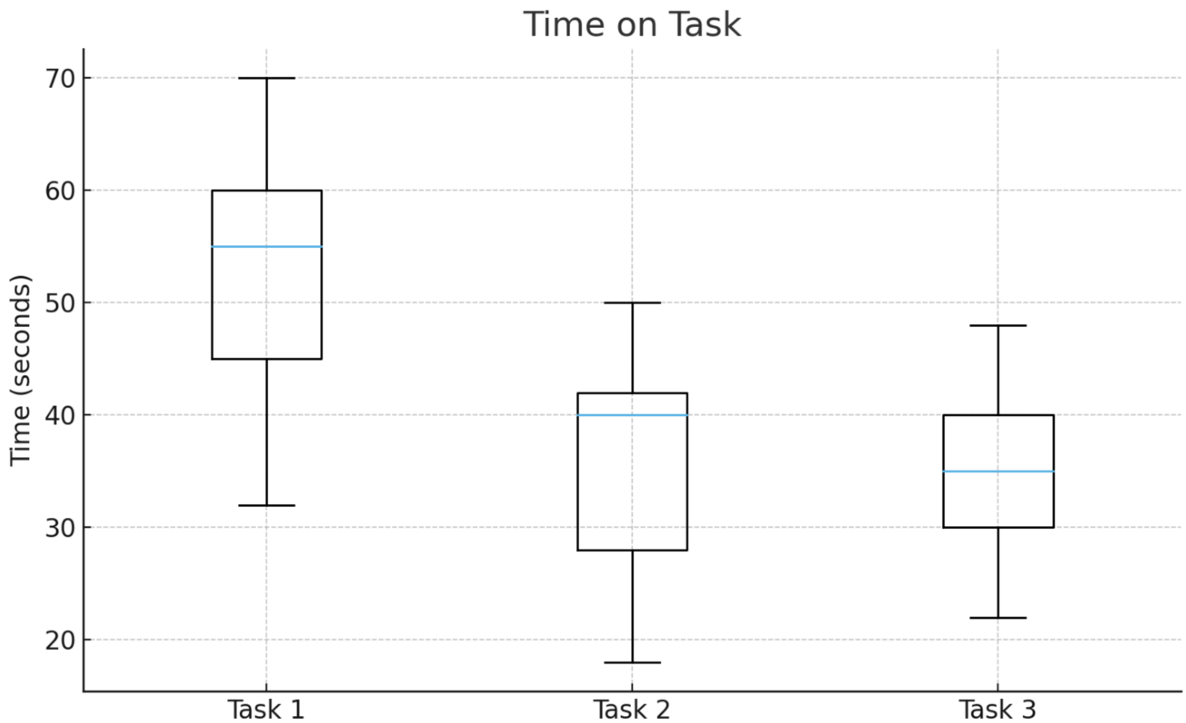
Expected impact: Reduces cognitive load, especially for less experienced users

These recommendations focus on changes that would provide the strongest gains in overall usability without disrupting existing workflows.

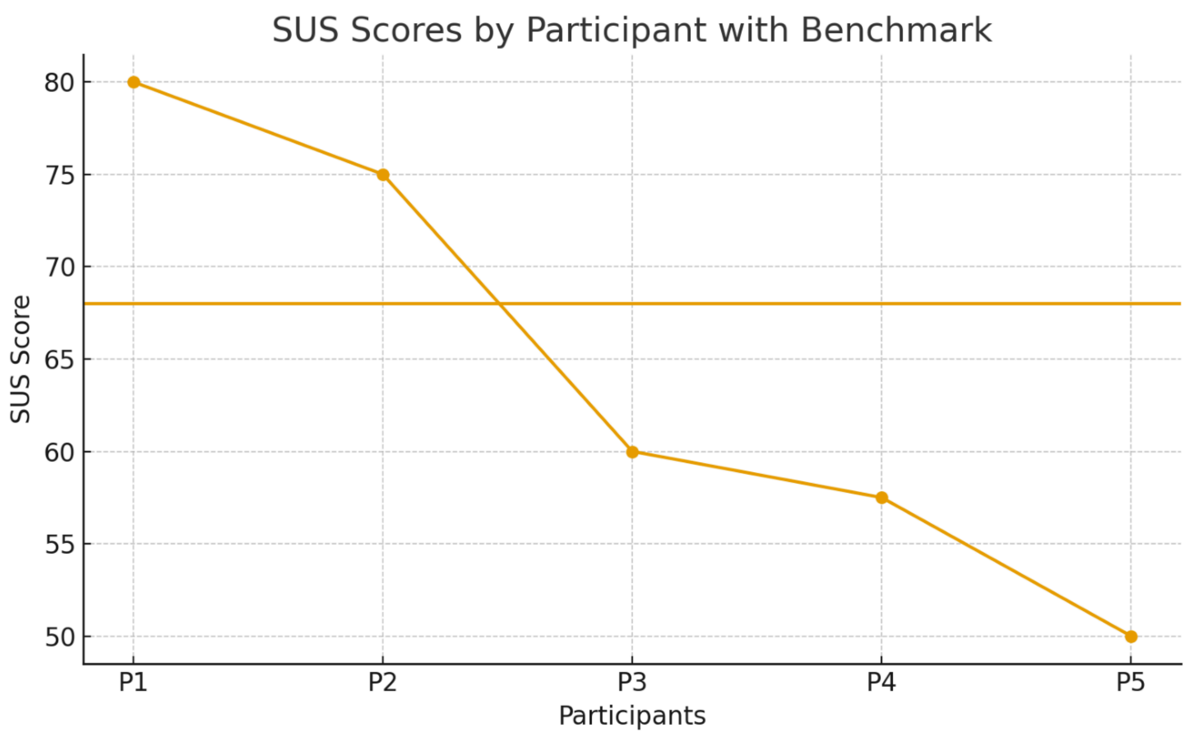
Visual Findings and Screenshots

Note: Detailed screenshots showing each identified usability problem are provided in the Key Findings and Analysis section. The visualizations below present quantitative performance data across all participants and tasks.





Time on Task
Time (seconds)
70
60
50
40
30
20
Task 1
Task 2
Task 3



SUS Scores by Participant with Benchmark
SUS Score
80
75
70
65
60
55
50
P1
P2
P3
P4
P5
Participants

7. Conclusion

This usability study revealed that the Starbucks mobile app performs reliably for common tasks such as placing an order or locating a nearby store, especially for confident mobile users. However, challenges with customization visibility, reload flow clarity, and map navigation created noticeable drops in Effectiveness, Efficiency, and Satisfaction for several participants. The final SUS score of 64.5 reinforces these findings, placing the overall usability of the app just below the industry average. Addressing the identified problem areas with targeted interface improvements would significantly enhance the experience for both new and returning users, creating a smoother, more intuitive workflow across the app.

8. Appendix

Appendix A: Raw Quantitative Data

Task Completion Summary

Participant	Task 1	Task 2	Task 3
P1	Completed	Completed	Completed
P2	Completed	Completed	Completed
P3	Assisted	Not Completed	Completed
P4	Completed	Not Completed	Completed
P5	Assisted	Not Completed	Completed

Time on Task (Seconds)

Participant	Task 1	Task 2	Task 3
P1	32	18	22
P2	45	28	35
P3	60	40	30
P4	55	42	48
P5	70	50	40

Appendix B: Qualitative Observation Notes

Error Notes

- P1: brief pause on customization.
- P2: backed out once during customization.
- P3: multiple scroll overs; cancelled reload attempt.
- P4: confusion in reload screen; selected wrong option once.
- P5: tapped wrong store icon; cancelled reload action.

1) Raw Qualitative Observation Notes

Verbal Feedback

- “The customize button is tiny.”
- “I’m not sure what this wants me to do.”
- “I don’t want to reload by accident.”
- “Where do I change the syrup?”
- “Let me zoom in… okay I see it now.”

Behavioural Cues

- Long pauses on customization screen.
- Repeated scrolling and backtracking.
- Hovering over reload options without selecting.
- Zooming in and out on map view.
- Occasional mis-taps or selection errors.

Emotional Responses

- Mild frustration during reload task (P3, P4, P5).
- Light confusion during map scanning (P2, P4).
- No major frustration during drink ordering.

Appendix C: SUS Scores

Participant SUS Scores

Participant	SUS Score
P1	80.0
P2	75.0
P3	60.0

P4	57.5
P5	50.0

Final System SUS Score

64.5 (below benchmark 68, indicating moderate usability with areas for improvement)